# Research project plan

## Shared team plan

synthetic-team-demo

Factory 2.6.0 | 2026-10-25T01:43:11+00:00

**DRAFT FOR REVIEW - NOT PREPARATION APPROVAL**

**40 of 43 responses recorded | 3 unresolved | 4 kits**

Configuration SHA-256: 8bd81fc53369f2262cf6e85bf7db1fa4
1f1d5bfba25e3ed9cfc26bf8ab0edba9

Workflow SHA-256: af46be98e2ed11136d637310c2336ce52cf0ca3067c9ec6d11c3112038500f8e

Public context, shared questions and role directory, checklist coverage, and explicitly selected coordinator sections. Private role setup, other answers and human decision history are excluded.

## Purpose and shared context

Illustrative project: neighborhood shade and summer surface temperature. This is a synthetic software demonstration, not a live study or an approved research finding. Compare matched observation units, retain original records and document uncertainty before drawing conclusions.

## Team and responsibility map

| Kit | Role | Human operator |
|---|---|---|
| hub | HUB | Synthetic PI |
| ingestor | INGESTOR | Synthetic intake operator |
| spoke-a | SPOKE | Synthetic researcher A |
| spoke-b | SPOKE | Synthetic researcher B |

## Research questions and assignments

### Q1

How does canopy cover relate to matched surface-temperature observations?

Lead: spoke-a | Support: spoke-b

Source: Illustrative project brief, research question Q1.

## Recorded project decisions

## Project Charter

### Purpose

Estimate how observed surface temperature varies with canopy cover within a defined study area.

### Deliverables

A documented source inventory, reproducible analysis, uncertainty summary and reviewed evidence package.

### Deadlines

Six-week demonstration schedule: source review in week 1, design review in week 2, analysis in weeks 3-4, challenge review in week 5 and final package in week 6.

### Constraints

Use licensed, authorized data. Surface temperature does not directly establish personal heat exposure. This demonstration contains no actual results.

### Completion conditions

Complete the source audit, analysis checks and separate human review gates; retain the exact methods, artifacts and limits.

## Research Logic

### Questions

Q1: How does canopy cover relate to matched surface-temperature observations?

### Research modes

Observational comparison with sensitivity analysis; causal claims require additional identification assumptions.

### Alternatives

Season, acquisition time, land cover, spatial scale and measurement error can explain apparent differences.

### Discriminating evidence

Check comparable acquisition conditions, variable definitions and predeclared alternative specifications.

## Method And Verification Plan

### Selection rationale

Use STATISTICAL_ANALYSIS to define the estimand, sampling unit, outcome and uncertainty before selecting an analysis. Spatial dependence needs an appropriate workstream-specific review.

### Verification

Audit missing values, units and joins; preserve original data; compare the primary result with documented sensitivity checks. A peer reviewer checks reasoning and reproducibility.

### Ethics and control documents

Use only authorized records. Check licensing and disclosure limits. No personal identifiers are required for this illustrative design.

# Provenance Policy

### Depth

Retain original source identifiers, versions, retrieval dates, processing steps and parent artifacts.

### Source rules

Use source documentation to establish units, temporal coverage and allowed use. Preserve conflicting or missing metadata.

### Failure memory

Record failed checks, corrections and rejected assumptions with their evidence; do not silently replace prior history.

### Handoff requirements

Send exact artifact versions, methods, limitations, human review and supplied conversation evidence through the approved route.

# Automation And Infrastructure Plan

### Storage

Selected arrangement: an institutional shared repository with recipient-specific folders. This example does not establish any live account access.

### Runtime adapter

Local Python tools and the operator-selected AI workspace; confirm actual file access before operational use.

### Checkpoint cadence

Save after completed review milestones and before handoff or interruption. Each shared kit has one active checkpoint writer.

### Transport

Human-mediated upload with exact package hashes and receiving-operator acknowledgment. A local file is not evidence of remote delivery.

### Approved services

Repository storage is planned. Automatic messaging is not selected. The Hub is the sole publisher of approved Library additions and corrections.

# Intake coverage and remaining questions

Question recorded means an ask event exists. Response recorded means a current answer remains in the checklist. These checks expose omissions; they do not prove that a human read or understood the plan.

Checklist protocol: ra-architect-workflow/2

| Stage | Required question | Asked | Response |
|---|---|---|---|
| 1 | Would you like to ask anything about the system or setup process before we begin? | Yes | Recorded |
| 1 | Which AI workspace will conduct this setup? | Yes | Recorded |
| 1 | Which verified Python 3.11-or-newer invocation will generate the kits? | Yes | Recorded |
| 2 | What project brief should this team use? | Yes | Recorded |
| 2 | What project name and stable project ID do you want? | Yes | Recorded |
| 2 | Who is the accountable project authority? | Yes | Recorded |
| 2 | Do you want person-based or role-based kit names? | Yes | Recorded |
| 3 | Do you confirm the displayed research questions with their source references? | Yes | Recorded |
| 3 | Which competing explanations must the research consider? | Yes | Recorded |
| 3 | What outputs will establish that the project is complete? | Yes | Recorded |
| 4 | Who will operate each proposed kit? | Yes | Recorded |
| 4 | What relevant expertise does each proposed operator bring? | Yes | Recorded |
| 4 | How many hours per week can each operator commit? | Yes | Recorded |
| 4 | How much review work can each reviewer realistically accept? | Yes | Recorded |
| 4 | What is the project due date? | Yes | Recorded |
| 4 | What intermediate milestones must the plan meet? | Yes | Recorded |
| 5 | Which evidence sources are available for the proposed work? | Yes | Recorded |
| 5 | What evidence could challenge the proposed explanations? | Yes | Recorded |
| 5 | Which ethical,access or regulatory constraints apply? | Yes | Recorded |
| 6 | Do you confirm the proposed research-mode mapping for each question? | Yes | Recorded |
| 7 | Do you confirm the displayed per-workstream methods and their reasons? | Yes | Recorded |
| 7 | What verification will each workstream require? | Yes | Recorded |
| 8 | Do you confirm the displayed lead/support assignments? | Yes | Recorded |
| 8 | Do you confirm the displayed workstream dependencies? | Yes | Recorded |
| 8 | Who is the single checkpoint writer for each shared kit? | Yes | Recorded |

| Stage | Required question | Asked | Response |
|---|---|---|---|
| 9 | Do the displayed workstream examples capture the consequential decisions requiring review? | Yes | Recorded |
| 9 | Who performs Gate-1 review for each workstream? | Yes | Recorded |
| 9 | Who performs Gate-2 review for each workstream? | Yes | Recorded |
| 9 | What reviewer overlaps or independence limits must be recorded? | Yes | Recorded |
| 10 | Do you confirm the proposed evidence-provenance requirements? | Yes | Recorded |
| 10 | What checkpoint cadence should the team use? | Yes | Recorded |
| 10 | Do you confirm the proposed handoff and transcript requirements? | Yes | Recorded |
| 11 | What model profiles are actually available to each operator? | Yes | Recorded |
| 11 | Do you accept the dated per-role model recommendations and their stated tradeoffs? | Yes | Recorded |
| 11 | Who will recheck model recommendations when the stated triggers occur? | Yes | Recorded |
| 12 | Which storage location will the team use? | Yes | Recorded |
| 12 | Which actual transport route will move files between people? | Yes | Recorded |
| 12 | Which required permissions have actually been configured? | Yes | Recorded |
| 12 | Which optional services do you want configured? | Yes | Recorded |
| 12 | Where should the new project folders be created? | Yes | Recorded |
| 13 | What corrections do you want to the complete architecture proposal? | Yes | Missing |
| 13 | Are any consequential decisions still unresolved in the displayed proposal? | Yes | Missing |
| 14 | Do you approve this exact displayed configuration for kit preparation? | Yes | Missing |

## How to use this plan

This document is generated from the recorded configuration. It does not grant access or research approval. Current checkpoint records retain later progress and closed work. Revise the configuration and regenerate this view; do not maintain a separately edited PDF.

Each operator also receives their own SETUP_HANDOFF.md with setup owners and pending actions. Actual setup completion, operator acknowledgment and subsequent corrections belong in current checkpoint records. The Hub alone publishes approved Library additions and corrections.